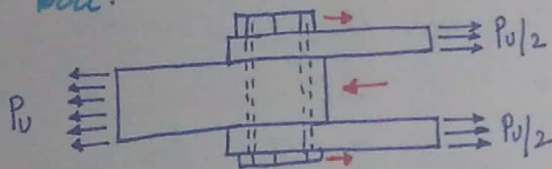


Failure of Single bolt fitting.

This is one of the simplest methods of joining two structural members together. Such a joint can transmit relatively large loads and yet the joint is easily disconnected. Such a joint can fail by bolt shear, bolt bending and bolt in tension.

Failure by bolt shear:-

In the fig shown, if a pull P' is placed on the fittings it tends to shear the bolt.



Let P_u be the ultimate load on the fitting, This force P_u must be resisted by the shear strength at two sections.

hence $P_u = 2 \cdot A \cdot F_{su}$, where F_{su} = Ultimate shearing stress for bolt-material.

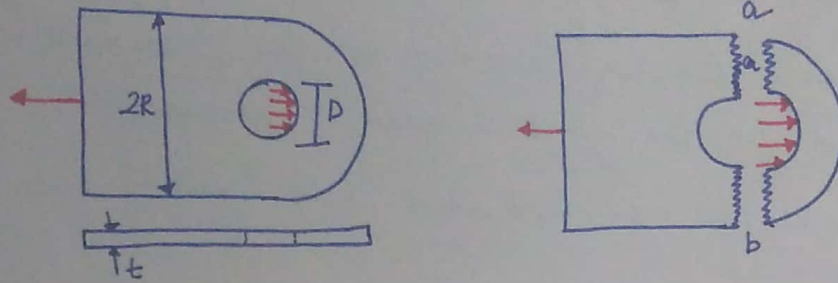
Failure by bolt bending:-

The main concern with bolt bending is that once the bolt has bent permanently but has not failed, Removal of the bolt becomes a problem.

BOLT & LUG STRENGTH ANALYSIS

Failure in Tension:-

Fig shows how a fitting plate can pull apart due to tension stresses on a section through the centerline of the bolt hole.



Hence, the stresses at points 'a' and 'b' are given as,

$$P_{u(t)} = F_{tu}(2R - D)t.$$

Where F_{tu} is the Ultimate tensile strength of the plate material.

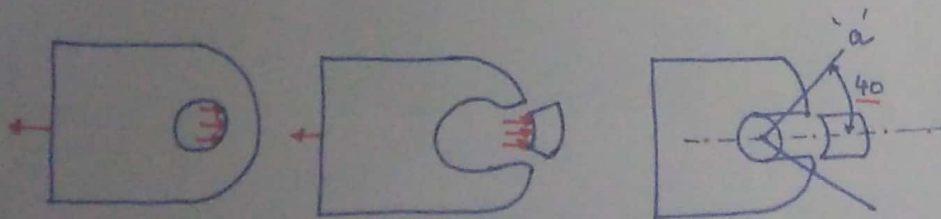
We should also note that due to the presence of the hole there will be stress concentrations, and we assume that stress concentration is uniform over the cross section, hence an increase in the factor of safety has to be made (25%) to overcome this effect.

Failure by shear tear out:-

In the shown fig, load 'P' causes a portion to tear out, we can write the equation as,

$$P_{u(\text{shear out})} = F_{su} \cdot A_s, \text{ where } F_{su} = \text{Ultimate shearing strength of the plate.}$$

$A_s = \text{Shear out Area.}$



The shearing area becomes the line 'a' times the thickness 't'. And not the edge length times thickness.

Failure by bearing on plate:-

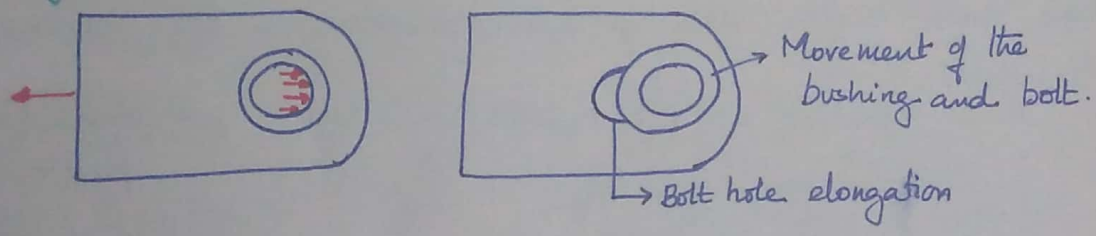
In the shown figure, the pull 'P' causes the bolt to press against the bushing wall, which in turn applies the pressure against the plate wall. If the pressure is high, then, the plate material will displace allowing the bolt and the bushing to move. Hence, we can write,

$$P_b = F_{br} \cdot D \cdot t, \text{ where, } F_{br} = \text{allowable bearing stress}$$

$$D = \text{diameter of bushing}$$

$$t = \text{plate thickness.}$$

An increase in fos of 50% is required, when designing a bushing / bearing joint.



Failure by bearing of Bolt Bushing:-

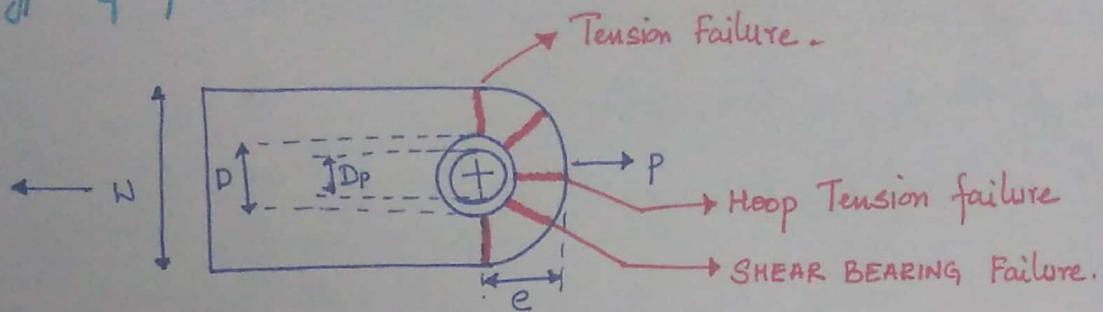
A bushing is pressed into a plate hole, and thus it is a tight fit. Unlike a bolt. A bolt fitting is usually designed for removable and refit, hence a certain tolerance, b/w the bolt and bushing. Hence, when there is a reversal of load, there is a shock on the fitting. And also due to the tolerance, causes the bolt to rotate causing wear b/w bolt and the fitting.

The equation used is similar to the previous section.

The tolerance also causes shock and vibration and hence an increase in the factor of safety (fos) should be used.

LOG STRENGTH ANALYSIS UNDER AXIAL LOADING

This section deals with a Combined shear out-bearing failure. Fig below shows the lug pin combination and types of failure



(a) TENSION ACROSS THE NET SECTION :-

As discussed, there will be stress concentrations around the hole and hence the former equations in the previous will be changed.

$$P_u = K_t \cdot F_{tu} \cdot A_t, \text{ where, } K_t = \text{stress concentration factor}$$

A_t = Net tension area.

(b) Shear out Bearing strength :-

Failure due to shear out and bearing is governed by the equation

$$P_{b,u} = K_{b,u} F_{tu} A_{br}$$

and similarly, $P_{b,y} = K_{b,y} F_{ty} A_{br}$, where $K_{b,u}$ and

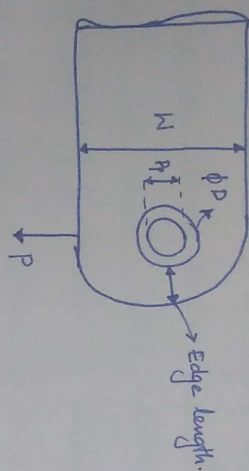
$K_{b,y}$ are shear bearing efficiency factor for yield strength and ultimate strength.

(c) Bushing yield :-

The allowable bearing load on bushing is given by.

$$P_{b,y} = 1.85 F_{cy} \cdot A_{br}, \text{ where } F_{cy} \text{ is compressible yield strength of bushing material,}$$

In this case, the lug & a fitting is subjected to only transverse loading. Transversely loaded lugs and bushings are checked in the same manner as axially loaded lugs. But the transversely loaded lug is more under stress than axially loaded lugs.



The equation governing a transversely loaded lug after is given

$$P_{LU} = K_{LU} A_{br} F_{LU} \text{, where } K_{LU} \text{ and } F_{LU} \text{ Ultimate and}$$

$$\phi \quad P_{LU} = K_{LU} A_{br} F_{LU} \text{ yield efficiency factor.}$$

Design STRENGTH ANALYSIS UNDER OBLIQUE LOADS:-

When fitting lugs are subjected to oblique loads, then the load is resolved and the following interaction equation is used,

$$P_a^{1/b} + P_{LU}^{1/b} = 1.$$

$$\text{and Margin of Safety (FOS)} = \frac{1}{(P_a^{1/b} + P_{LU}^{1/b})^{0.525}} - 1$$

P_a = Axial component of Ultimate load divided by P_u ,

$$\text{where } P_u = K_{LU} A_{br} F_{LU} \text{ or } P_{LU} = K_{LU} A_{br} F_{LU}$$

wherever is smaller.

P_{LU} = Transverse component of the Ultimate load divided by P_u ,

$$\text{where } P_u = K_{LU} A_{br} F_{LU} \text{ (from above section)}$$

Welding
An expensive method is attached to vessels.

ADVANTAGES OF BOLTED JOINTS :-

(6)

- (a) Easy assembly and Disassembly.
- (b) Can take tension loads.
- (c) Bolts don't change properties of the structure unlike welds.
- (d) Bolting is easy unlike rivets & welds where skilled labour is required.
- (e) Bolts aren't sensitive to condition of parent metal.

DISADVANTAGES OF BOLTED JOINTS

- (a) Requires access on both sides of the joint.
- (b) Can get loose over time.
- (c) Requires holes, hence stress concentrations go up.
- (d) Pre-loading can be tough to measure.
- (e) Corrosion b/w bolt and parent metal.

These are used to connect together structural members

Rivets are best used to bear shear loads, also used in steel trusses and bridges. Riveted joints when subjected to vibrations and impact tend to be more reliable when compared to vibrations and impact. These type of joints are more permanent joints when compared to boltings.

Rivets used in aluminium have better applications, where the weldability is very poor.

Rivets generally are designed to take up only shear loads and not tension, and this the clamping loads in Rivets is less when compared to bolts.

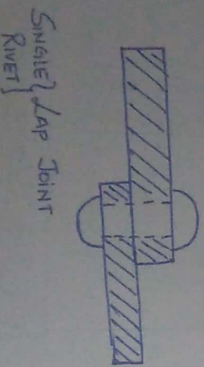
Rivets also lead to a fail safe type of design. And Rivets can also bear high fatigue loads, and quality of Rivet joints can be easily inspected.

Types of Riveted Joints:-

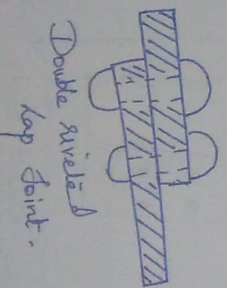
- (a) Lap joints (b) Butt joints.

Lap Joints:- The edges of the plate are joined together by an overlap of edges. There are two types of lap joints.

- ① Single riveted lap joint.
- ② Double riveted lap joint.
- ③ Triple riveted lap joint. etc.



Single Riveted Lap Joint



Double Riveted Lap Joint

Butt Joints:

Butt joints are made by touching the two plates without an overlap and a cover plate is placed either on one side or both the sides of the members.

Atleast 2 rows of rivets are required on each side.

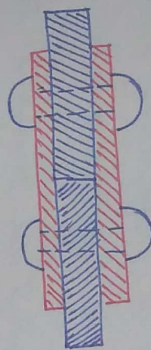
Types of Butt joints are:-

- ① Single riveted
- ② Double riveted
- ③ Triple riveted.

Rivets



Covering plate



Single cover plate:-

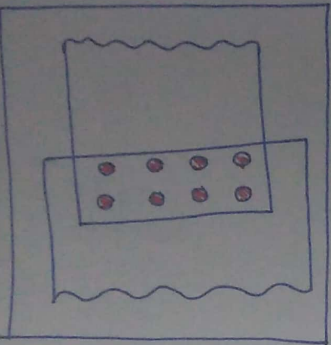
Parent Metal

Double cover plate:-

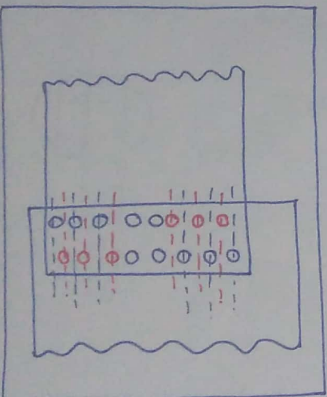
In Addition to the above the following are the other types of Riveted joints:-

- ② Chain Riveted joints.
- ③ Zig Zag Riveted joints
- ④ Diamond Riveted Joints.

- ① Chain Riveted joints.
- ② Zig Zag Riveted joints.

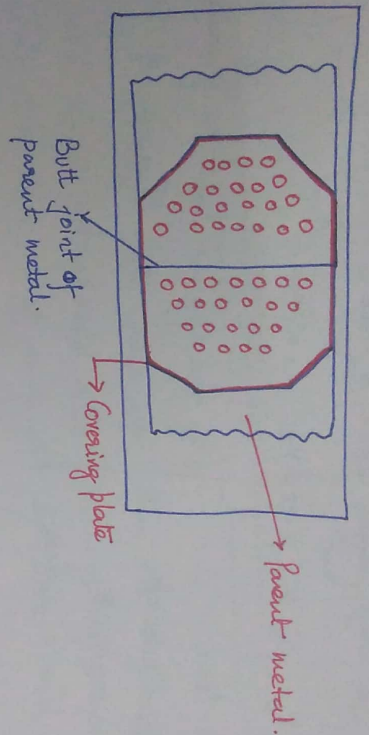


Chain Riveted joints.



Zig Zag Riveted joints.

This type of Riveting is only used in Butt joints.



Failure of a Riveted Joint:

A riveted joint may fail in the following ways.

- (a) Tearing of plate b/w rivet hole & edge of plate.
- (b) Tearing of plates b/w rivets.
- (c) Shearing of rivet.
- (d) Crushing/Bearing of rivet.
- (e) Crushing of plate.

(a) Failure by tearing of plate b/w rivet hole & edge of plate:-

If the distance b/w centre of rivet & nearest edge of plate is not sufficient, the tearing will take place, when loaded. This failure may be avoided if

$$m = 1.5d$$

m = margin & distance b/w centre of hole & edge)

d = diameter of rivet.

⑧. Failure due to tearing of plate b/w rivets in row is

The safe tensile load per pitch length is given by.

$P_t = \text{Safe tensile stress} \times \text{Area of plate per pitch length.}$

$$P_t = \sigma_t \times (p-d) \times t \quad , \text{ where,}$$

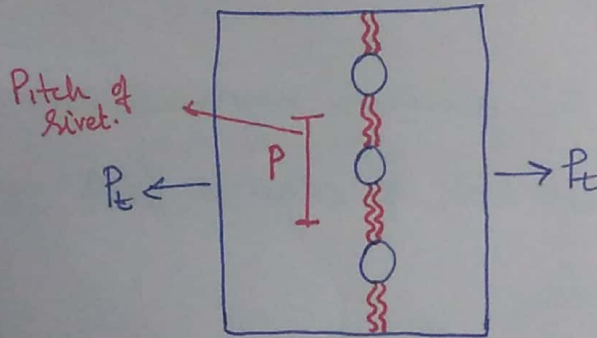
$P_t = \text{tensile load,}$

$d = \text{diameter of rivet}$

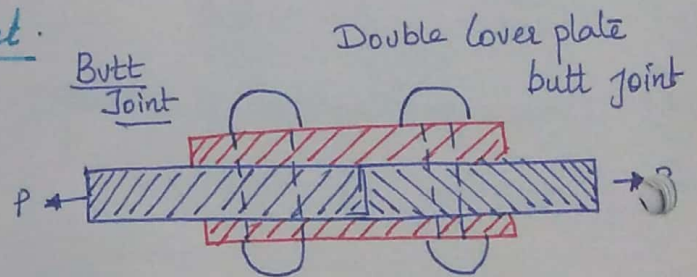
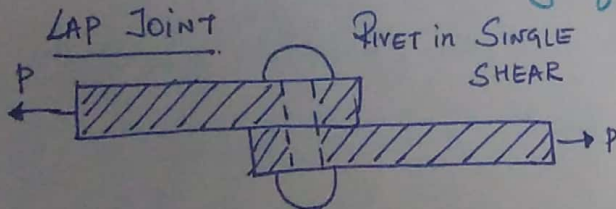
$t = \text{thickness of plate,}$

$\sigma_t = \text{Safe tensile stress.}$

if the applied load is greater than P_t then the tearing of plate will occur b/w rivets.



⑨. Failure due to shearing of rivet.



The Gross section resisting shear when rivet is in single shear is $\frac{\pi}{4} d^2$ and when the rivet is in double shear it is $2 \times \frac{\pi}{4} d^2$.

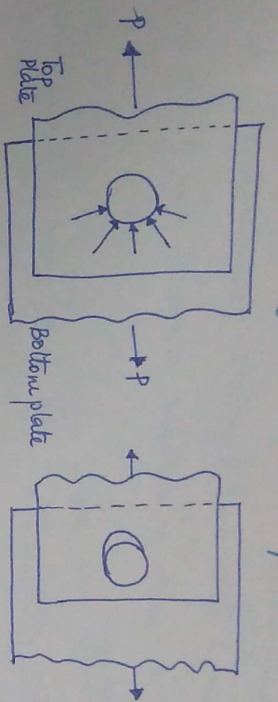
$$\begin{aligned} (1) \text{ Safe load: } P_s &= \frac{\pi}{4} d^2 \times \tau \text{ for single riveted lap joint.} \\ &= 2 \times \frac{\pi}{4} d^2 \times \tau \text{ for double riveted lap joint.} \\ &= 3 \times \frac{\pi}{4} d^2 \times \tau \text{ for triple riveted lap joint.} \end{aligned} \quad \left[\begin{array}{l} \text{Also for} \\ \text{butt joints} \end{array} \right]$$

✗

This formula is also used for butt joints with single cover plate.

Failure due to crushing of rivet or plate :-

(11)



Assuming the top part is weaker than the bottom plate, when the load is applied, the crushing stress/bearing stress is applied b/w the rivet and the top plate.

If the load is large, then the plate or the rivet can get crushed.

$$P_b = \sigma_b \times \text{area} = \sigma_b \times d \times t, \text{ where } d, \text{ is diameter}$$

t , is thickness of rivet-plate.

Strength of a riveted joint :-

The maximum force a riveted joint can withstand is called, strength of the joint.

The strength of the joint for rivets in single shear is

$$P_s = n \times Z \times \frac{\pi}{4} d^2, \text{ where } n \text{ is no. of rivets.}$$

$$\text{and for double shear} = n \times Z \times 2 \times \frac{\pi}{4} d^2$$

and for crushing strength is $P_b = n \times \sigma_b \times d \times t$, where n is no. of rivets.

WELDED JOINTS :-

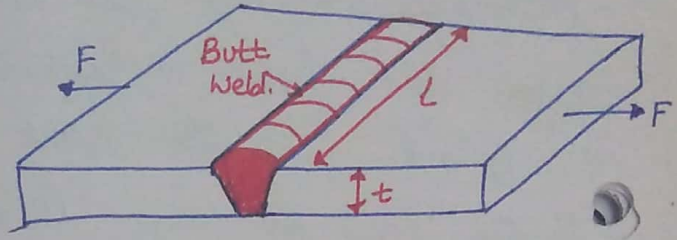
Process of joining two metal joints permanently by the method of fusion with or without pressure and a filler material is called welding:-

Types of Welded joints are:- (a) Butt weld (b) Lap weld.

(a) Butt Weld joint :- Edges of the members are touching each other and are welded together.

The tensile force F is given by.

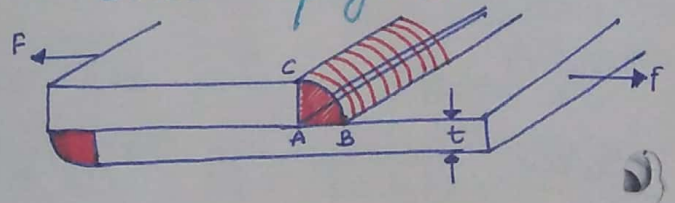
$$F = \sigma_t \times L \times t$$



The above mentioned equation can also be used in the calculation of the strength of the butt weld joint.

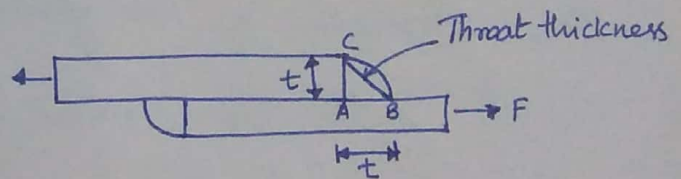
(b) filled or Lap Weld joint :- When two overlapping members are joined by weld the joint is called lap joint.

$$\text{Throat thickness} = t \times \frac{1}{\sqrt{2}} \\ = 0.707t.$$



Tensile strength of double filled lap joint is given by.

$$\begin{aligned} F &= \sigma_t \times \text{Area of double filled weld.} \\ &= \sigma_t \times 2 \times \text{length} \times \text{throat thickness} \\ &= \sigma_t \times 2 \times L \times t(0.707) \end{aligned}$$



Similarly for a single filled weld.

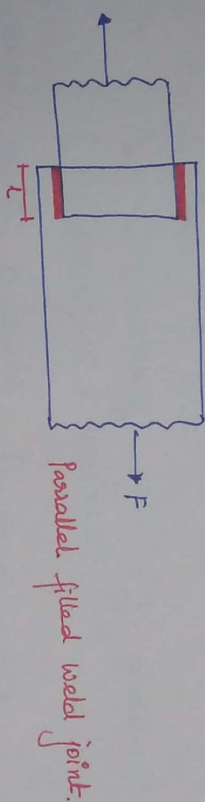
$$F = \sigma_t \times L \times 0.707t.$$

Types of fillet weld joints:

- ① Single fillet lap joint.
- ② Double fillet lap joint
- ③ Parallel fillet lap joint.

Single fillet lap joint and double fillet lap joint, already discussed in previous section.

Parallel fillet lap joint:



Area of the parallel fillet weld joint is given by:

$$= (L \times \text{throat thickness}) \times 2$$

$$= (L \times 0.707t) \times 2 = 1.414 \times L \times t$$

Analysis of Unsymmetrical welded Sections:

Moment of Top weld, $= F_1 \times a$
 $= S \times L_1 \times a$

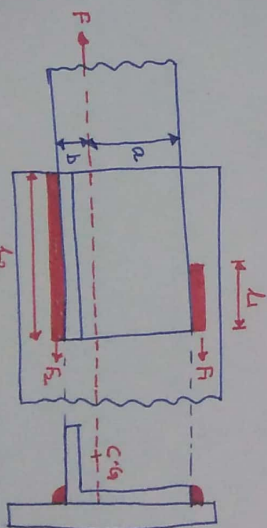
Moment of bottom weld $= F_2 \times b$
 $= S \times L_2 \times b$

From $\sum M_x = 0$, $S \times L_1 \times a = S \times L_2 \times b$

$$\frac{a}{b} = \frac{L_2}{L_1} \text{ or } L_2 = \frac{a L_1}{b}$$

Total length of weld $= L_1 + L_2 = L$

$$L = L_1 + L_2 = L_1 + \frac{a L_1}{b}$$



- Load F is acting through C.G.

$F = F_1 + F_2$ for equilibrium.

$F_1 = S \times L_1$
 $F_2 = S \times L_2$ where S is the resistance of weld per unit length.

$$L = l_1 + l_2 = l_1 + \frac{a \cdot l_1}{b} = \frac{b l_1 + a l_1}{b} = \frac{l_1 (a+b)}{b} \quad \text{--- (1)}$$

from this we have a relation b/w the total length and l_1 and l_2

$$\text{Similarly, } L = \frac{l_2 (a+b)}{a} \quad \text{--- (2)}$$

Hence,

$$F_1 = S \times L_1$$

rearranging equation (1), we get,

$$F_1 = S \times \frac{b L}{(a+b)}, \quad F_1 = \frac{F b}{(a+b)} \quad \left[\text{we know } S \times L = F \right]$$

$$\text{Similarly } F_2 = \frac{a \times F}{(a+b)} \quad //.$$

~~Therefore~~ Axial loads in terms of shear stress

$$F = \tau \times (l_1 + l_2) \times 0.707 t$$

which is, Axial load = shear stress $\times$ Area of weld.

Ans:-

15

① A double riveted lap joint is made b/w 15mm thick plates. The rivet diameter & pitch are 85mm & 75mm. If the ultimate stress are 400 mpa in tension and 320 mpa in shear and 640 mpa in crushing. Find the minimum force which will rupture the joint.

If the joint is subjected to a factor of safety of 4, find the actual stresses developed in the plates.

Sol ② Failure due to tearing of plate:-

$$P_t = \sigma_t (P-d) \times t$$

$$P_t = (75-25) \times 15 \times 400 = 3 \times 10^5 \text{ N.}$$

③ Ultimate shearing resistance of rivets

$$P_s = \frac{\pi}{4} d^2 \times \tau \times n \quad \text{where } n=2 \text{ for double riveted joint}$$

$$P_s = \frac{\pi}{4} \times (25)^2 \times 320 \times 2$$

$$P_s = 3.14 \times 10^5 \text{ N.}$$

④ Crushing resistance, $P_{cr} = \sigma_b \times d \times t \times n$

$$P_b = 640 \times 25 \times 75 \times 2 = 4.8 \times 10^5 \text{ N.}$$

From the above, we can say that minimum force for the joint to fail is $3 \times 10^5 \text{ N.}$

Actual stresses produced are:

Since $fos = 4$, $3 \times 10^5 \text{ N} / 4 = 7.5 \times 10^4 \text{ N.}$

$$\text{① } \sigma_t = \frac{P_t}{(P-d) \times t} = \frac{75 \times 10^3 \text{ N}}{(75-25) \times 15} = 100 \frac{\text{N}}{\text{mm}^2}$$

$$\text{② } \tau = \frac{P_s}{\frac{\pi}{4} \times 25^2 \times 2} = \frac{75 \times 10^3 \text{ N}}{981.25} = 76.43 \frac{\text{N}}{\text{mm}^2}$$

$$\text{③ } \sigma_b = \frac{P_b}{d \times t \times n} = \frac{75 \times 10^3}{25 \times 75 \times 2}$$

$$\sigma_b = 100 \frac{\text{N}}{\text{mm}^2}$$

Design of a riveted joint:

(16)

Design of riveted joint is to find the following quantities.

(a) Diameter of rivets:- $d = 1.6 \sqrt{t}$ when d and t are in cm

$d = 6 \sqrt{t}$ when d & t are in mm.

For plates with thickness greater than 8mm

(b) Pitch of rivets $p = \frac{P \cdot d}{P}$ where P = pitch of rivet.

$$\text{or } \frac{P \cdot d}{P} = \sigma_t (P - d) \times t = P_t \text{ or } P_s$$

(c) No. of rivets required.

$$n = \frac{P}{\text{least rivet value for shearing or bearing}}, P = \text{Force.}$$

(d) Thickness of cover plate (butt joint)

$t_1 = 1.25 t$ for butt joint with single cover plate

$t_1 = 0.635 t$ " " " double " "

As a criterion, pitch should be between 2.5 and 3.0 times d .

Example:

Two plates 10mm thick are joined by a single riveted lap joint. The plates are subjected to a load of 200kN. If the permissible tensile, shear and bearing stresses are 120N/mm², 100N/mm² and 160N/mm², determine.

- (a) Diameter of rivets (b) Pitch of rivet (c) Number of rivets
(d) efficiency of rivets.

Sol:- Given $t = 10\text{mm}$

Lap joint, Single shear.

$$P = 200 \text{ kN.}$$

$$\sigma_t = 120 \text{ N/mm}^2$$

$$\text{and } \sigma_b = 160 \text{ N/mm}^2.$$

$$\tau = 100 \text{ N/mm}^2$$

Diameter of rivet,

$$d = 6\sqrt{t} = 18.97 \text{ approx. } 19 \text{ mm. for } 100 \text{ mm thickness.}$$

(17)

⑥ Pitch of rivet,
 $P \text{ or } P_c = \sigma_t (P-d) \times t$ as efficiency is not known.

$$P_s = \frac{\pi}{4} d^2 \times \tau = \frac{\pi}{4} \times 19^2 \times 100 = 28.3 \times 10^3 \text{ N. } \left\{ \begin{array}{l} \text{Taking the least value} \\ \text{among the two.} \end{array} \right.$$

$$P_b = d \times t \times \sigma_b = 19 \times 10 \times 160 = 30.4 \times 10^3 \text{ N.}$$

$$P = 28.3 \times 10^3 \text{ N.}$$

$$28.3 \times 10^3 = 120 (P-19) \times 10, \quad P = 42.6 \text{ mm.}$$

Pitch should be b/w 2.5 and 3.0 times d.

$\frac{P}{d} = \frac{42.6}{19} = 2.25$, hence, we have to multiply d by 2.5 to maintain criteria.

$$P = 19 \times 2.5 = 47.5 \text{ mm}$$

⑦ Number of rivets:- $n = \frac{P}{\text{least value for shear \& bearing}}$

$$\text{least rivet is } 28.3 \times 10^3 \text{ N} \quad n = \frac{20 \times 10^3}{28.3 \times 10^3} = 7.06 \text{ or } 8 \text{ rivets.}$$

⑧ Efficiency of this joint:-

$$\eta = \frac{\text{least value of } P_s, P_b \& P_c}{\sigma_t \cdot P \cdot t} = \frac{28.3 \times 10^3}{120 \times 47.5 \times 10}$$

$$\eta = 49.74 \%$$

Example: ③

A double riveted double cover butt joint in plates 20mm thick is made with 25mm diameter rivets at 100mm pitch.

The permissible stresses are:

$$\sigma_t = 120 \text{ mpa} \quad \tau = 100 \text{ mpa} \quad \sigma_b = 150 \text{ mpa}.$$

Find the efficiency of joint, taking the strength of the rivet in double shear as twice than that of single shear.

Sol: ① $P_t = (p-d) \times t \times \sigma_t = (100-25) \times 20 \times 120 = 18 \times 10^4 \text{ N}.$ Least of the three is $15 \times 10^4 \text{ N}.$

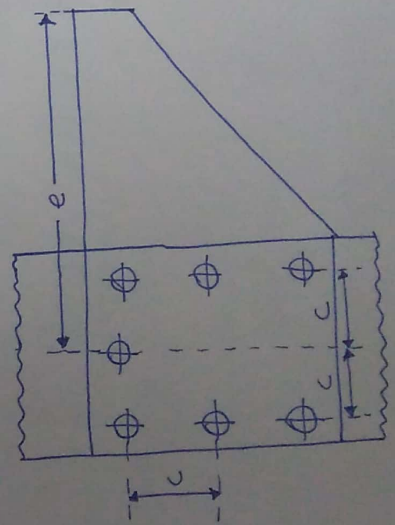
$P_s = n \times 2 \times \frac{\pi}{4} d^2 \times \tau = 2 \times 2 \times \frac{\pi}{4} 25^2 \times 100 = 19.6 \times 10^4 \text{ N}.$

$P_b = n \times d \times t \times \sigma_b = 2 \times 25 \times 20 \times 150 = 15 \times 10^4 \text{ N}.$

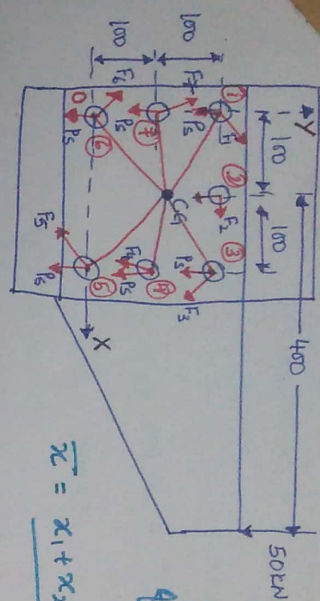
② efficiency of the joint = $\frac{\text{strength of riveted joint}}{\text{strength of un-riveted solid plate}}.$

$$= \frac{15 \times 10^4 \text{ N}}{P \times t \times \sigma_t} = \frac{15 \times 10^4 \text{ N}}{100 \times 20 \times 120} = 62.5\%.$$

Example ④ An eccentrically loaded lap riveted joint is to be designed for a steel bracket as shown.



The bracket plate is 25mm thick. All rivets are of the same size. Load on bracket is 50kN. Rivet spacing is $C = 100 \text{ mm}$. $e = 400 \text{ mm}$. $\sigma_b = 120 \text{ mpa}$ $\tau = 65 \text{ mpa}$. Determine size of the rivets to be used for the joint.



Calculate the centre of gravity of the rivet system.

(19)

$$\bar{x} = \frac{x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_4}{n}$$

$$\bar{y} = \frac{y_1 + y_2 + y_3 + y_4 + y_5 + y_6 + y_4}{n}$$

$x_1, x_2, \dots$ distance from Oy
 $y_1, y_2, \dots$ distance from Ox

$$\bar{x} = \frac{0 + 100 + 200 + 200 + 200 + 0 + 0}{7} = 100 \text{ mm.}$$

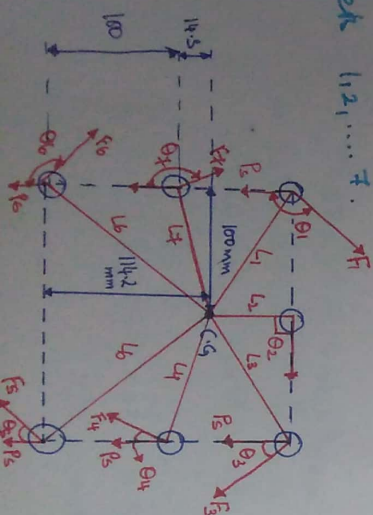
$$\bar{y} = \frac{200 + 200 + 200 + 100 + 0 + 0 + 100}{7} = \frac{800}{7} = 114.28 \text{ mm.}$$

As shown in fig above.

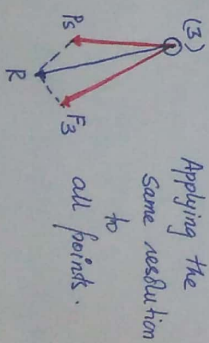
Shear load on each rivet: $P = \frac{50 \times 10^3 \text{ N}}{7} = 7143 \text{ N.}$ acting vertically downwards at each rivet.

$$\text{Moment produced} = P \times e = 50 \times 10^3 \times 400 = 2 \times 10^7 \text{ N-mm.}$$

As seen in the image/figure above, $F_1, F_2, \dots, F_7$ be secondary shear loads at a 1st distance $L_1, L_2, \dots, L_7$ from the C.G. for the rivets $1, 2, \dots, 7$.



at point 3



$$L_1 = L_3 = \sqrt{100^2 + (200 - 114.3)^2} = 131.7 \text{ mm}$$

$$L_2 = 200 - 114.3 = 85.7 \text{ mm}$$

$$L_4 = L_7 = \sqrt{100^2 + (114.3 - 100)^2} = 101 \text{ mm}$$

$$L_5 = L_6 = \sqrt{100^2 + 114.3^2} = 152 \text{ mm}$$

In order to find shear load the following assumptions are made.

(1) The secondary shear load is proportional to the radial distance from C.G.

(2) The direction of the secondary shear is $\perp$ to the line joining the centre of rivet to centre of gravity.

$F_1 \propto L_1$ and $F_2 \propto L_2 \dots \dots \dots$ So on.

$$\frac{F_1}{L_1} = \frac{F_2}{L_2} = \frac{F_3}{L_3} \dots \dots \dots \text{So on.} \quad \therefore F_2 = F_1 \frac{L_2}{L_1} \text{ and } F_3 = F_1 \frac{L_3}{L_1}$$

Now, equating the turning moment due to eccentricity to resisting moment of the rivets.

$$\Rightarrow P \times e = F_1 \times L_1 + F_2 \times L_2 + F_3 \times L_3 \dots \dots \dots$$

$$P \times e = \frac{F_1}{L_1} [L_1^2 + L_2^2 + L_3^2 + L_4^2 + L_5^2 + L_6^2 + L_7^2]$$

$$P \times e = \frac{F_1}{131.7} [2(L_1)^2 + L_2^2 + 2(L_4)^2 + 2(L_5)^2], \quad L_1 = L_3, \quad L_4 = L_7, \quad L_5 = L_6$$

$$50 \times 10^3 \times 400 = \frac{F_1}{131.7} [108645] \quad \therefore 20 \times 10^6 \times 131.7 = F_1 [108645]$$

$$\therefore \boxed{F_1 = 24244 \text{ N.}} \quad F_2 = F_1 \frac{L_2}{L_1}, \quad F_2 = 15776 \text{ N, Similarly,}$$

$$F_3 = 24244 \text{ N} \quad F_5 = 27981 \text{ N} \quad F_7 = 18593 \text{ N.}$$

$$F_4 = 18593 \text{ N} \quad F_6 = 27981 \text{ N}$$

calculating angles.

$$\theta_{37} \cos \theta_3 = \frac{100}{L_3} = 0.76$$

$$\cos \theta_4 = \frac{100}{L_4} = 0.99$$

$$\cos \theta_5 = \frac{100}{L_5} = 0.658$$

Calculating resultant force.

$$R_3 = \sqrt{R_5^2 + F_3^2 + 2 R_5 F_3 \cos \theta_3}$$

$$= \sqrt{(7143)^2 + (24244)^2 + 2 \times 7143 \times 24244 (0.76)}$$

$$= 30033 \text{ N.}$$

Similarly calculating other results.

$$R_4 = 85684 \text{ N}$$

$$R_5 = 33121 \text{ N}$$

From the resultant shear load, on rivet S the diameter d' can be calculated.

$$R_5 = \frac{\pi}{4} d'^2 \times \tau > \frac{33121}{\tau \times \frac{\pi}{4}} = d'^2, \quad \sqrt{\frac{33121}{\tau \times \frac{\pi}{4}}} = d'$$

$$d = 85.5 \text{ mm}$$

Now, we check if the diameter will withstand the crushing stress.

$$\text{Crushing stress or bearing stress} = \frac{R_5}{\text{drt}} = \frac{\text{Max load}}{\text{Area}}$$

$$= \frac{33121}{85.5 \times 85} = 51.96 \text{ N/mm}^2 \text{ which is less than}$$

$\sigma_b = 120 \text{ N/mm}^2$, hence, we conclude design is safe.

Examples on welded joints:

- ① Two steel plates 10mm wide & 125mm thick are to be joined by double lap weld joint. Find the length of the weld if the maximum tensile stress is not to exceed 75 N/mm^2 and max tensile load carried by the plates is 100kN.

Sol. Using equation for double lap weld joint,

$$F = \sigma_t \times 2 \times \text{length of weld} \times \text{throat thickness}$$

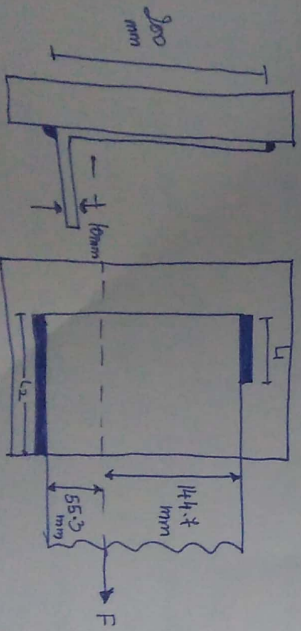
$$F = \sigma_t \times 2 \times L \times 1.414 \times 12.5$$

$$100 \times 10^3 = 75 \times 2 \times L \times 1.414 \times 12.5$$

$$L = 75.4 \text{ cm or } 754 \text{ mm.}$$

Ans 754mm

- ② A 800mm x 150mm x 10mm angle, carrying a load of 200kN is to be welded to a steel plate by fillet welds, as shown in fig. Find the lengths of the weld at the top and bottom if the allowable shear stress in the weld is 102.5 N/mm^2 . The distances b/w neutral axis and edges are 144.7mm and 55.3mm.



Equation is derived as follows.

$$F = \tau \times (L_1 + L_2) \times 0.707 t.$$

$$F = \tau \times (l_1 + l_2) \times 0.707 t.$$

$$l = l_1 + l_2$$

= total length of weld.

(23)

$$2000 \times 10^3 = 108.5 \times (l_1 + l_2) \times 0.707 (10)$$

$$l_1 + l_2 = 276 \text{ mm} = l$$

$$\text{We know } l_1 = \frac{b \times l}{a + b} = \frac{55.3 \times 276}{(14.7 + 55.3)} = 26.3 \text{ mm}$$

$$\text{and hence } l_2 = 199.7 \text{ mm} //$$

Example 3 A welded lap joint is provided to connect two tie bars $15 \text{ cm} \times 1 \text{ cm}$ as shown in fig. The working stress in tie bar is 150 N/mm^2 . Investigate the design if the size of the fillet weld is 8 mm . The safe stress for the weld may be taken as 102 N/mm^2 .

Sol: Nature of weld = Lap joint.

$$\text{Area of tie bar} = 15 \text{ cm}^2 = 1500 \text{ mm}^2.$$

Now load the tie bar can resist.

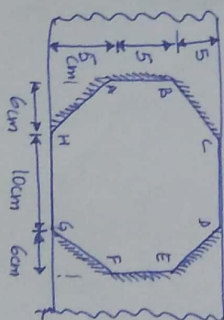
$$\begin{aligned} &= \text{Working stress} \times \text{Area} \\ &= 150 \text{ N/mm}^2 \times 1500 \text{ mm}^2 = 225 \times 10^4 \text{ N.} \end{aligned}$$

$$\text{Throat thickness} = 0.707 \times t = 0.707 \times 8 = 5.65 \text{ mm.}$$

$$\text{Safe stress} = 102 \text{ N/mm}^2.$$

$$\text{Total length of weld} = \text{length } (AB + BC + CD + ED + EF + FG + GH + HA)$$

$$\text{Calculating: Total length of weld} = 412.6 \text{ mm.}$$



Strength of weld = Safe stress $\times$ Area of weld.

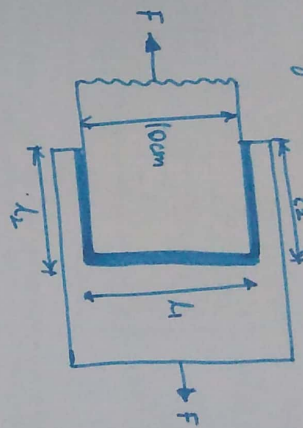
$$= 102 \times (\text{Total length of weld} \times \text{Throat thickness})$$

$$= 102 \times (412.6 \times 5.65)$$

$$= 237918.5 \text{ N.}$$

Since, Safe ~~stress~~^{load} is greater than working load the joint is safe. with an f.o.s of 1.05.

plate 10cm wide and 1.20cm thick is joined with another plate by a single filled lap weld and a double parallel filled weld as shown. The maximum tensile and shear stresses are 75 N/mm^2 respectively. Find the length of each parallel weld if the joint is subjected to a total load of 9000 N.



load carried by

Single filled lap joint is

$$F_1 = 0.707 \times \sigma_t \times l_1 \times t$$

$$= 0.707 \times 75 \times 100 \times 12$$

$$= 63630 \text{ N}$$

$$F_2 = 1.414 \times l_2 \times t \times \tau$$

$$= 1.414 \times l_2 \times 12 \times 55 = 933.24 \times l_2$$

$$\text{Total load} = F = F_1 + F_2$$

$$9000 = 63630 + 933.24 \times l_2$$

Solving, $l_2 = 28.25 \text{ mm}$.